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Inventor(s)	LaTorraca; Gary John et al.

Variable throat jet venturi

Abstract

A variable throat jet venturi for delivering ventilation gas to a patient includes a jet nozzle, a deformable throat body arranged to receive ventilation gas output by the jet nozzle and defining a gas inlet and a gas outlet, and a housing containing the deformable throat body. The housing may define an entrainment opening which is open to ambient air and a pilot pressure port for pressurizing a plenum between an outer wall of the deformable throat body and an inner wall of the housing. A pilot pressure line may be fluidly coupled to the pilot pressure port. A controller may be programmed to energize the pilot pressure line to constrict the deformable throat body during an exhalation phase of positive end-expiratory pressure (PEEP) therapy.

Inventors: LaTorraca; Gary John (Batesville, IN), Westfall; Tom (Batesville, IN)

Applicant: HILL-ROM SERVICES PTE. LTD. (Ot, SG)

Family ID: 80218509

Assignee: HILL-ROM SERVICES PTE. LTD. (Ot, SG)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority to U.S. Provisional Application Ser. No. 63/146,850 filed Feb. 8, 2021, the disclosure of which is incorporated herein by reference.

STATEMENT RE: FEDERALLY SPONSORED RESEARCH/DEVELOPMENT

(1) Not Applicable

BACKGROUND

(2) Non-invasive open ventilation (NIOV), in which the patient's airway remains open rather than being sealed by a mask, is of great advantage to patients with chronic obstructive pulmonary disease (COPD) and other respiratory conditions due to its minimal interference with activities of daily living. However, because the patient can freely exhale, there is no easy way to generate positive end-expiratory pressure (PEEP) for NIOV patients and doing so typically requires a high flow rate of gas to the patient, typically air and/or oxygen. The gas consumed generating PEEP is wasted, as is the energy to produce it. This results in larger, heavier devices and shorter battery run time. The required flow rate of gas can be reduced using pneumatic (e.g. poppet) valves or eliminated altogether using electrically operated valves. However, such valves are physically large, requiring the undesirable placement of a bulky element close to the patient's face, in addition to adding complexity and weight to the device and introducing dead space volume that increases carbon dioxide rebreathing.

BRIEF SUMMARY

(3) The present disclosure contemplates various systems and methods for overcoming the above drawbacks accompanying the related art. One aspect of the embodiments of the present disclosure is a variable throat jet venturi for delivering ventilation gas to a patient. The variable throat jet venturi may comprise a jet nozzle, a deformable throat body arranged to receive ventilation gas output by the jet nozzle and defining a gas inlet and a gas outlet, and a housing containing the deformable throat body. The housing may define an entrainment opening which is open to ambient air and a pilot pressure port for pressurizing a plenum between an outer wall of the deformable throat body and an inner wall of the housing.

(4) With the plenum in a first pressurization state, the variable throat jet venturi may achieve a shutoff pressure $P_{sub.shutoff}$ at the gas outlet of at least 55 cmH₂O at a jet nozzle pressure $P_{sub.n}$ of 10.5 psig and a jet nozzle flow $V'_{sub.n}$ equal to or less than 30 slpm. With the plenum in a second pressurization state, the variable throat jet venturi may achieve a gas outlet pressure $P_{sub.aw}$ of 10 cmH₂O at a jet nozzle flow $V'_{sub.n}$ of less than 10 slpm, e.g. less than 5 slpm. With the plenum in the first pressurization state, a ratio $A_{sub.t}/A_{sub.n}$ of a cross-sectional area of the deformable throat body to a cross-sectional area of the jet nozzle may be between 20 and 30. With the plenum in the second pressurization state, the ratio $A_{sub.t}/A_{sub.n}$ may be between 2.0 and 5.0.

(5) The deformable throat body may be arranged to receive the ventilation gas output by the jet nozzle through the entrainment opening of the housing.

(6) Another aspect of the embodiments of the present disclosure is a patient ventilation interface. The patient ventilation interface may comprise the above variable throat jet venturi and a nasal coupler for fluidly coupling the gas outlet of the deformable throat body to a nostril of the patient. The nasal coupler may comprise a nasal pillow.

(7) Another aspect of the embodiments of the present disclosure is a non-invasive ventilation system. The non-invasive ventilation system may comprise the above patient ventilation interface and a pilot pressure line fluidly coupled to the pilot pressure port. The non-invasive ventilation system may comprise a controller programmed to energize the pilot pressure line to constrict the deformable throat body during an exhalation phase of positive end-expiratory pressure (PEEP) therapy. The non-invasive ventilation system may comprise a multi-lumen tube having a ventilation gas lumen terminating in the nozzle and a pilot pressure lumen in fluid communication with the pilot pressure line.

(8) Another aspect of the embodiments of the present disclosure is a method of varying a ratio between a throat diameter and a jet nozzle diameter of a variable throat jet venturi for delivering ventilation gas to a patient. The method may comprise providing a deformable throat body arranged to receive ventilation gas output by a jet nozzle and ambient air entrained via an entrainment opening, the deformable throat body defining a gas inlet and a gas outlet. The method may further comprise pressurizing a plenum between an outer wall of the throat body and an inner wall of a housing containing the deformable throat body to constrict the deformable throat body.

(9) The pressurizing of the plenum may include energizing a pilot pressure line fluidly coupled to a pilot pressure port defined by the housing. The energizing of the pilot pressure line may be performed during an exhalation phase of positive end-expiratory pressure (PEEP) therapy.

(10) The pressurizing of the plenum may comprise pressurizing the plenum from a first pressurization state, in which the variable throat jet venturi achieves a shutoff pressure $P_{sub.shutoff}$ at the gas outlet of at least 55 cmH₂O at a jet nozzle pressure $P_{sub.n}$ of 10.5 psig and a jet nozzle flow $V'_{sub.n}$ equal to or less than 30 slpm, to a second pressurization state, in which the variable throat jet venturi achieves a gas outlet pressure $P_{sub.aw}$ of 10 cmH₂O at a jet nozzle flow $V'_{sub.n}$ of less than 10 slpm, e.g. less than 5 slpm. With the plenum in the first pressurization state, a ratio $A_{sub.t}/A_{sub.n}$ of a cross-sectional area of the deformable throat body to a cross-sectional area of the jet nozzle may be between 20 and 30. With the plenum in the second pressurization state, the ratio $A_{sub.t}/A_{sub.n}$ may be between 2.0 and 5.0.

(11) Another aspect of the embodiments of the present disclosure is a variable throat jet venturi for delivering ventilation gas to a patient. The variable throat jet venturi may comprise a jet nozzle, a deformable throat body arranged to receive ventilation gas output by the jet nozzle and ambient air entrained via an entrainment opening and defining a gas inlet and a gas outlet, and a housing containing the deformable throat body. The housing may define a pilot pressure port for pressurizing a plenum between an outer wall of the deformable throat body and an inner wall of the housing.

(12) With the plenum in a first pressurization state, the variable throat jet venturi may achieve a shutoff pressure $P_{sub.shutoff}$ at the gas outlet of at least 55 cmH₂O at a jet nozzle pressure $P_{sub.n}$ of 10.5 psig and a jet nozzle flow $V'_{sub.n}$ equal to or less than 30 slpm. With the plenum in a second pressurization state, the variable throat jet venturi may achieve a gas outlet pressure $P_{sub.aw}$ of 10 cmH₂O at a jet nozzle flow $V'_{sub.n}$ of less than 10 slpm, e.g. less than 5 slpm. With the plenum in the first pressurization state, a ratio $A_{sub.t}/A_{sub.n}$ of a cross-sectional area of the deformable throat body to a cross-sectional area of the jet nozzle may be between 20 and 30. With the plenum in the second pressurization state, the ratio $A_{sub.t}/A_{sub.n}$ may be between 2.0 and 5.0.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features and advantages of the various embodiments disclosed herein will be better understood with respect to the following description and drawings, in which like numbers

refer to like parts throughout, and in which:

- (2) FIG. 1 shows an exemplary non-invasive ventilation system according to an embodiment of the present disclosure, with a variable throat jet venturi of the system shown in cutaway;
- (3) FIG. 2A is a cross-sectional view of the variable throat jet venturi with a plenum thereof in a first pressurization state;
- (4) FIG. 2B is a cross-sectional view of the variable throat jet venturi with the plenum in a second pressurization state;
- (5) FIG. 3A is a perspective view of the variable throat jet venturi;
- (6) FIG. 3B is an exploded view of the variable throat jet venturi;
- (7) FIG. 3C is an exploded sectional view of the variable throat jet venturi;
- (8) FIG. 4 shows four exemplary candidate venturi geometries in relaxed and constricted states corresponding to first and second plenum pressurization states; and
- (9) FIG. 5 graphically depicts performance characteristics of an exemplary variable throat jet venturi.

DETAILED DESCRIPTION

(10) The present disclosure encompasses various embodiments of a variable throat jet venturi for delivering ventilation gas to a patient, along with systems and methods for varying a ratio between a throat diameter and a jet nozzle diameter thereof. The detailed description set forth below in connection with the appended drawings is intended as a description of several currently contemplated embodiments and is not intended to represent the only form in which the disclosed invention may be developed or utilized. The description sets forth the functions and features in connection with the illustrated embodiments. It is to be understood, however, that the same or equivalent functions may be accomplished by different embodiments that are also intended to be encompassed within the scope of the present disclosure. It is further understood that the use of relational terms such as first and second and the like are used solely to distinguish one from another entity without necessarily requiring or implying any actual such relationship or order between such entities.

(11) FIG. 1 shows an exemplary non-invasive ventilation system **10** according to an embodiment of the present disclosure, including a patient ventilation interface **20** that incorporates a variable throat jet venturi **100** for delivering ventilation gas to a patient. Due to the increased velocity of the ventilation gas at a constriction of the variable throat jet venturi **100**, there is a decrease in pressure that causes ambient air to be entrained via one or more entrainment ports **133**. By amplifying the ventilation gas output by a jet nozzle **110** in this way, the variable throat jet venturi **100** may serve as an efficient flow generator when providing ventilation therapy to the patient. Meanwhile, so that it may also serve as an efficient PEEP generator, the variable throat jet venturi **100** may include a deformable throat body **120** to serve as the constriction, with a housing **130** of the variable throat jet venturi **100** defining a pilot pressure port **135** for pressurizing a plenum **140** surrounding the deformable throat body **120** (e.g. via a feed and bleed circuit). By selectively pressurizing the plenum **140**, the plenum **140** can be transitioned between a first pressurization state for maximizing airflow to the patient (e.g. during inhalation) and a second pressurization state in which the deformable throat body **120** is more constricted. In the latter state, the reduced cross-sectional area of the deformable throat body **120** may significantly reduce the required nozzle flow for achieving a desired output pressure, making it possible to efficiently generate PEEP.

(12) As shown by way of example in FIG. 1, the patient ventilation interface **20** in which the variable throat jet venturi **100** is integrated may be a nasal interface having a pair of nasal couplers **24** (e.g. nasal pillows) for fluidly coupling a gas outlet of the deformable throat body **120** to the nostrils of the patient. Examples of such a nasal interface are described in U.S. Pat. No. 9,132,250, entitled METHODS, SYSTEMS AND DEVICES FOR NON-INVASIVE VENTILATION INCLUDING A NON-SEALING VENTILATION INTERFACE WITH AN ENTRAINMENT PORT AND/OR PRESSURE FEATURE, U.S. Pat. No. 9,675,774, entitled METHODS,

SYSTEMS AND DEVICES FOR NON-INVASIVE OPEN VENTILATION WITH GAS DELIVERY NOZZLES IN FREE SPACE, U.S. Pat. No. 9,962,512, entitled METHODS, SYSTEMS AND DEVICES FOR NON-INVASIVE VENTILATION INCLUDING A NON-SEALING VENTILATION INTERFACE WITH A FREE SPACE NOZZLE FEATURE, and U.S. Pat. No. 10,792,449, entitled PATIENT INTERFACE WITH INTEGRATED JET PUMP, the entire contents of each of which are expressly incorporated by reference herein. For example, the variable throat jet venturi **100** may be disposed within or constitute a manifold assembly **22** of the patient ventilation interface **20**, with the jet nozzle **110** integrated therewith or attachable to an end thereof. The manifold assembly **22** may be configured to fit between the patient's nose and upper lip and may serve to direct exit flow from the variable throat jet venturi **100** to the patient's nostrils.

(13) With reference to FIG. **1**, it is contemplated that the variable throat jet venturi **100** may be integrated in or otherwise provided at one side of the manifold assembly **22** only, with the exit flow thereof being directed simultaneously to both of the patient's nostrils as shown. Along these lines, as indicated above, the manifold assembly **22** of the patient ventilation interface **20** may be outfitted with the spaced, identically configured pair of the nasal couplers **24** (e.g. nasal pillows) engageable to and placeable into fluid communication with respective ones of the patient's nostrils, and the manifold assembly **22** may place the variable throat jet venturi **100** into fluid communication with both such nasal couplers **24** by a single flow passage. Alternatively, the manifold **22** may define separate flow passages for each of the patient's left and right nostrils, in which case the variable throat jet venturi **100** and some or all of its features described herein may be duplicated for each side of the manifold **22**.

(14) In order to provide for the pressurization of the plenum **140** as described above, the non-invasive ventilation system **10** may include a pilot pressure line **30** fluidly coupled to the pilot pressure port **135**. A controller **40** may be programmed to energize the pilot pressure line **30** to constrict the deformable throat body **120** during an exhalation phase of PEEP therapy. In the illustrated example, the non-invasive ventilation system **10** includes a multi-lumen tube **50** having a ventilation gas lumen **52** terminating in the jet nozzle **110** and a pilot pressure lumen **54** in fluid communication with the pilot pressure line **30**. The ventilation gas lumen **52** may receive ventilation gas from a ventilator or an oxygen concentrator, for example. The pilot pressure line **30** may extend from the multi-lumen tube **50** forward past the jet nozzle **110** to the pilot pressure port **135** of the variable throat jet venturi **100**.

(15) Additional lumens of the multi-lumen tube **50** may include, for example, a low-pressure gas lumen for oxygen (which may terminate in a low-pressure jet nozzle outlet port near the jet nozzle **110**), a pressure sensing lumen (which may extend farther downstream to terminate nearer to the patient's nostril, such as at the base of the nasal coupler **24** closest the variable throat jet venturi **100**, for example), lumens for medicaments, etc. However, it is contemplated that in a preferred implementation, a second tube **50** may be provided which routes one or more of these lumens to the other side of the manifold assembly **22** (i.e., the side opposite that having the variable throat jet venturi **100** integrated therein). Such a pair of tubes **50** may branch upstream from a single multi-lumen tube using a wye connector such as that described in U.S. Pat. No. 10,792,449, incorporated by reference above. In FIG. **1**, for example, a second tube **50** is shown having, as its only lumen, a pressure sensing lumen **56** that extends to the base of one of the nasal couplers **24** in order to desirably measure pressure downstream of the variable throat jet venturi **100**. In this example, the first tube **50** will have no pressure sensing lumen. Along these lines, it is contemplated that any of the above-described lumens may be provided individually or in some prescribed combination at one or both sides of the interface **20** using the first tube **50** with or without the second tube **50**.

(16) In a case where the non-invasive ventilation system **10** may include only a single multi-lumen tube **50** extending to the patient ventilation interface **20**, it is also contemplated that the multi-lumen tube **50** may interface with the manifold **22** at a central position equidistant from the nasal couplers **24**. For example, the multi-lumen tube **50** may connect to the manifold **22** between the

nasal couplers **24** on the bottom or the front of the manifold **22** as viewed in FIG. **1**. In such case, this same central area, near the bases of the nasal couplers **24**, may house a single variable throat jet venturi **100** that is fed by the multi-lumen tube **50** and whose output flow is directed simultaneously to both nasal couplers **24**. As another possibility, the disclosed variable throat jet venturi **100** may be embodied in a detachable connector that interfaces the multi-lumen tube **50** with a mask or other patient interface at any appropriate position (and in some cases may be universally usable with multiple different patient interfaces). Examples of such connectors that can be equipped with the disclosed variable throat jet venturi **100** are the adaptors disclosed in U.S. Pat. No. 10,307,552, entitled JET PUMP ADAPTOR FOR VENTILATION SYSTEM, the entire contents of which are expressly incorporated by reference herein.

(17) The controller **40** may be a standalone device dedicated to energizing the pilot pressure line **30** during PEEP therapy (e.g. based on sensor input indicative of an exhalation phase of a patient's breathing) or may be a controller of a ventilator or oxygen concentrator, for example. In this regard, exemplary ventilators and oxygen concentrators that may be used with the disclosed embodiments include, in addition to those incorporated by reference above, those described in U.S. Pat. No. 10,369,320, entitled MODULAR VENTILATION SYSTEM, U.S. Patent Application Pub. No. 2019/0307981, entitled MODULAR VENTILATION SYSTEM, and U.S. patent application Ser. No. 16/874,472, filed May 14, 2020 and entitled O₂ CONCENTRATOR WITH SIEVE BED BYPASS AND CONTROL METHOD THEREOF," the entire contents of each of which are expressly incorporated herein by reference. The controller **40** may energize the pilot pressure line **30** by controlling the pressure in the pilot pressure lumen **54** of the multi-lumen tube **50**, for example, by controlling a valve of a pilot pressure output port of a ventilator, oxygen concentrator, or other gas source that houses or is connected to the controller **40**.

(18) FIGS. **2A** and **2B** are cross-sectional views of the variable throat jet venturi **100** with the plenum **140** in the first pressurization state and the second pressurization state, respectively. FIGS. **3A-3C** are additional perspective, exploded and exploded sectional views of the variable throat jet venturi **100**. As can be seen in FIGS. **2A** and **2B**, when pressure increases in the plenum **140** (e.g. by activation of a pilot pressure line **30**, see FIG. **1**), the plenum **140** transitions from the first pressurization state (FIG. **2A**) to the second pressurization state (FIG. **2B**), with the buildup of pressure in the plenum **140** causing the deformable throat body **120** to become constricted. More specifically, a cross-sectional area $A_{sub.t}$ of the deformable throat body **120** (e.g. a smallest cross-sectional area corresponding to a diameter $D_{sub.t}$ in FIGS. **2A** and **2B**) decreases from the first pressurization state to the second pressurization state. Given a nozzle having a fixed cross-sectional area $A_{sub.n}$ (corresponding to a diameter $D_{sub.n}$), the ratio $A_{sub.t}/A_{sub.n}$ thus decreases, reducing the required nozzle flow for achieving PEEP.

(19) The deformable throat body **120** may define a gas inlet **122** and a gas outlet **124** and may have a generally tubular shape in its relaxed (e.g. as-molded) state as best shown in FIG. **3B**. Flanges **126** may be defined at either end. Referring back to FIGS. **2A** and **2B**, when the deformable throat body **120** is disposed within the housing **130**, the flanges **126** may function to position the deformable throat body **120** in a cavity defined therein. The flanges **126** may also serve to delineate a longitudinal extent of the plenum **140** between an outer wall **126** of the deformable throat body **120** and an inner wall **137** of the housing **130**. As the plenum **140** is pressurized as shown in FIG. **2B**, the pressure may cause the deformable throat body **120** to constrict about its center while the flanges **126** remain firmly against the inner wall **137** of the housing **130** (with or without the use of adhesive). Thereafter, as the plenum **140** is depressurized, the deformable throat body **120** may return to its relaxed state as shown in FIG. **2A**. The deformable throat body **120** may be a thermoplastic elastomer (TPE) or a thermoset produced by liquid injection molding (LIM) using liquid silicone rubber (LSR), for example.

(20) The housing **130** may be assembled from one or more pieces **132**, **134** as shown, which may be attached to each other by ultrasonic welding, for example. The pieces **132**, **134** of the housing

130 may similarly be made of a thermoplastic or thermoset but may typically (but not necessarily) have greater rigidity than the deformable throat body **120**. In the illustrated example, there is an entry piece **132** and an exit piece **134**. In more detail, as shown in FIGS. 2A-3C, the entry piece **132** of the housing **130** includes a generally frustoconical portion which flares outwardly away from the plenum **140** and defines an entrainment opening **136** open to ambient air. The exit piece **134** includes the inner wall **137** that defines the plenum **140**, and likewise includes a generally frustoconical portion which flares outwardly away from the plenum **140** and serves as a diffuser **138** that may provide the desired airflow and/or pressure to the patient, e.g. via the nasal couplers **24** (see FIG. 1). With the arrangement shown in FIG. 1, the deformable throat body **120** may receive the ventilation gas output by the jet nozzle **110** via the entrainment opening **136** of the housing **130**, in addition to ambient air which is drawn through the entrainment port **133** of the manifold assembly **22** and likewise introduced into the entrainment opening **136**. In this regard, as seen in FIG. 1, the entrainment opening **136** (and hence the deformable throat body **120**) fluidly communicates with both the jet nozzle **110** and entrainment port **133** of the manifold assembly **22**. The distal tip of the jet nozzle **110** may be either outside the entrainment opening **136** as shown or inside the entry piece **132** downstream of the entrainment opening **136**, closer to the gas inlet **122** of the deformable throat body **120**, with the entrainment of ambient air occurring around the periphery of the jet nozzle **110** in either case.

(21) In the implementation shown in FIG. 1 wherein the manifold assembly **22** includes the entrainment port **133** and the distal tip of the jet nozzle **110** is radially aligned with a portion of the entrainment port **133**, it is contemplated that the entrainment opening **136** may likewise be radially aligned with a portion of the entrainment port **133**, or located downstream therefrom. The same relative positioning options between the entrainment opening **136** and the entrainment port **133** are available if the distal tip of the jet nozzle **110** is disposed inside the entry piece **132** downstream of the entrainment opening **136**. While the entrainment port **133** is depicted in FIG. 1 as being formed in the manifold assembly **22**, it may alternatively be formed in a detachable nozzle assembly which contains the jet nozzle **110** and is rigidly or rotatably engaged to a corresponding end of an alternatively configured manifold assembly devoid of the entrainment port **133**. Along these lines, variants of the patient ventilation interface **20** shown in FIG. 1 are also contemplated wherein the entrainment opening **136** of the variable throat jet venturi **100** may be arranged to draw ambient air directly therein, as opposed to ambient air being channeled thereto from a separate entrainment port or opening such as the entrainment port **133**. In this instance, ambient air drawn directly into the entrainment opening **136** may flow about the periphery of the jet nozzle **110**, particularly if the distal tip thereof resides within the entry piece **132** downstream of the entrainment opening **136**. In any case, however, the deformable throat body **120** is arranged to receive, at its gas inlet **122**, the ventilation gas output by the jet nozzle **110** together with ambient air entrained into entrainment opening **136** either directly or after passing through the entrainment port **133**.

(22) The geometry of the variable throat jet venturi **100**, including the cross-sectional area of the nozzle **110** and in particular the geometry of the deformable throat body **120** when the plenum **140** is in the first and second pressurization states, may be selected to achieve desired performance characteristics. For purposes of illustration, maximum ventilator output capabilities may dictate a nozzle flow $V'_{sub.n} \leq 30$ slpm and a nozzle pressure $P_{sub.n} = 10.5$ psig, which, in turn, may limit the range of possible nozzle diameters. Using each of a plurality of possible nozzle diameters, candidate venturi geometries may be tested with the deformable throat body **120** in the relaxed state (plenum **140** in the first pressurization state as shown in FIG. 2A) to determine a shutoff pressure $P_{sub.shutoff}$ at the gas outlet **124** of the deformable throat body **120** and a maximum output flow $V'_{sub.aw-max}$ at the maximum nozzle pressure $P_{sub.n} = 10.5$. In this way, it can be confirmed whether the candidate venturi geometry meets desired $P_{sub.shutoff} - V'_{sub.aw-max}$ performance. The same candidate venturi geometries may then be tested with the deformable throat body **120** in the constricted state (plenum **140** in the second pressurization state as shown in FIG.

$V'.sub.aw-max = V'.sub.aw * P.sub.shutoff / (P.sub.shutoff - P.sub.aw)$. The ratio $X.sub.max$ of the maximum output flow $V'.sub.aw-max$ to the maximum nozzle flow $V'.sub.n$ can be calculated as follows: $X.sub.max = V'.sub.aw-max / V'.sub.n$.

(28) The performance characteristics of interest can then be tabulated from the combined data of the relaxed and constricted states of the deformable throat body **120** of each candidate venturi geometry, as summarized in Tables 3 and 4 below for nozzle diameters $D.sub.n=0.043$ inches and $D.sub.n=0.048$ inches, respectively:

(29) TABLE-US-00003 TABLE 3 RELAXED CONSTRICTED $V'.sub.n$ at $P.sub.n = P.sub.shutoff$ at $P.sub.n = V'.sub.aw-max$ at $P.sub.n = V'.sub.n$ at PEEP = # 10.5 psig 10.5 psig 10.5 psig
 $X.sub.max$ 10 mm H.sub.2O 1 19.6 35.0 85.8 4.4 3.9 2 19.5 39.0 96.4 4.9 4.1 3 19.9 36.0 93.3 4.7 3.8 4 19.4 39.0 105.2 5.4 4.5

(30) TABLE-US-00004 TABLE 4 RELAXED CONSTRICTED $V'.sub.n$ at $P.sub.n = P.sub.shutoff$ at $P.sub.n = V'.sub.aw-max$ at $P.sub.n = V'.sub.n$ at PEEP = # 10.5 psig 10.5 psig 10.5 psig
 $X.sub.max$ 10 mm H.sub.2O 1 28.1 40.0 79.8 2.8 4.1 2 28.9 47.5 87.6 3.0 4.5 3 27.8 41.0 77.4 2.8 4.3 4 28.2 47.0 90.8 3.2 4.3

(31) An exemplary venturi geometry that achieves desired performance characteristics is shown in Table 5, below, with unused data omitted, for a nozzle **110** having a nozzle diameter $D.sub.n=0.048$ inches (cross-sectional area $A.sub.n=0.0018$ in.^{sup.2}). Again, the deformable throat body **120** is assumed to receive the ventilation gas output by the jet nozzle **110** through the entrainment opening **136** of the housing **130** as shown in FIGS. 2A and 2B, with the length between the nozzle **110** and the gas inlet **122** of the deformable throat body **120** being 0.4 inches.

(32) TABLE-US-00005 TABLE 5 $D.sub.t$ $A.sub.t/$ $L.sub.t$ $L.sub.d$ $L.sub.t + L.sub.d$ $P.sub.n$ $V'.sub.n$ $P.sub.shutoff$ $P.sub.aw$ $V'.sub.aw$ # (in) $A.sub.n$ (in) (in) (in) (psig) (slpm) (cmH.sub.2O) (cmH.sub.2O) (slpm) 5.1 0.23 23 0.40 0.40 0.80 — — 10.0 — — 10.5 24.3 57.0 3.0 91.0 5.2 0.086 3 0.40 0.40 0.80 0.4 4.3 10.0 0.0 — 10.5 — — — —

(33) The tabulated performance characteristics of interest from the combined data of the relaxed and constricted states of the deformable throat body **120** of the above candidate venturi geometry #5 are summarized in Table 6 below:

(34) TABLE-US-00006 TABLE 6 RELAXED CONSTRICTED $V'.sub.n$ at $P.sub.n = P.sub.shutoff$ at $P.sub.n = V'.sub.aw-max$ at $P.sub.n = V'.sub.n$ at PEEP = # 10.5 psig 10.5 psig 10.5 psig
 $X.sub.max$ 10 mm H.sub.2O 5 24.3 57.0 96.1 4.0 4.3

(35) As can be seen from Tables 5 and 6 (and as graphically depicted in FIG. 5), a variable throat jet venturi **100** having candidate venturi geometry #5 achieves a shutoff pressure $P.sub.shutoff$ at the gas outlet **124** of 57.0 cmH.sub.2O (meeting target performance characteristics of at least 55 cmH.sub.2O) at a jet nozzle pressure $P.sub.n$ of 10.5 psig and a jet nozzle flow $V'.sub.n$ of 24.3 lpm (which is less than or equal to 30 slpm) while the plenum **140** is in a first pressurization state (uninflated) corresponding to the relaxed state of the deformable throat body **120**. A maximum output flow $V'.sub.aw-max=96.1$ is achieved. Meanwhile, when the plenum **140** is in a second pressurization state (inflated) corresponding to the constricted state of the deformable throat body **120**, the same variable throat jet venturi **100** achieves PEEP (gas outlet pressure $P.sub.aw=10$ cmH₂O) at a jet nozzle flow $V'.sub.n$ of 4.3 slpm. Thus, using a variable throat jet venturi **100** having a deformable throat body **120** as described herein, PEEP (e.g. gas outlet pressure $P.sub.aw=10$ cmH₂O) can be achieved at a jet nozzle flow $V'.sub.n$ of less than 10 slpm, which is less than half of the 20 slpm that is typically required with conventional NIOV devices and, in some cases, at a jet nozzle flow $V'.sub.n$ of less than 5 slpm, which is less than a quarter of typical requirements. It is worth noting, however, that energizing a pilot pressure line **30** as used in some implementations of the disclosed deformable throat body **120** may use some airflow, e.g. 1 slpm.

(36) Equivalent testing can be performed to select a venturi geometry that meets design constraints of any patient interface **20**, for example, one in which one or more entrainment openings **136** have a side-by-side relationship with the jet nozzle **110** and/or for different lengths between the nozzle

110 and the gas inlet **122** of the deformable throat body **120**. Along the same lines, the venturi geometry can be selected to meet different performance characteristics, including different maximum ventilator output capabilities other than nozzle flow $V'_{sub.n} \leq 30$ slpm and nozzle pressure $P_{sub.n} = 10.5$ psig, different PEEP other than 10 mmH_{sub.2O}, different target shutoff pressure $P_{sub.shutoff}$ and maximum output flow $V'_{sub.aw-max}$ at the maximum nozzle pressure $P_{sub.n}$, etc.

(37) The controller **40** of the non-invasive ventilation system **10** (which may be a controller of an oxygen concentrator or ventilator as noted above) may be implemented with a programmable integrated circuit device such as a microcontroller or control processor. Broadly, the device may receive certain inputs, and based upon those inputs, may generate certain outputs. The specific operations that are performed on the inputs may be programmed as instructions that are executed by the control processor. In this regard, the device may include an arithmetic/logic unit (ALU), various registers, and input/output ports. External memory such as EEPROM (electrically erasable/programmable read only memory) may be connected to the device for permanent storage and retrieval of program instructions, and there may also be an internal random access memory (RAM). Computer programs for implementing any of the disclosed functionality of the controller **40** may reside on such non-transitory program storage media, as well as on removable non-transitory program storage media such as a semiconductor memory (e.g. IC card), for example, in the case of providing an update to an existing device. Examples of program instructions stored on a program storage medium or computer-readable medium may include, in addition to code executable by a processor, state information for execution by programmable circuitry such as a field-programmable gate arrays (FPGA) or programmable logic device (PLD).

(38) In the above examples, a variable throat jet venturi **100** is implemented with a deformable throat body **120** whose cross-sectional area $A_{sub.t}$ is selectively changed relative to a fixed cross-sectional area $A_{sub.n}$ of a jet nozzle **110**. However, it is also contemplated that the cross-sectional area $A_{sub.n}$ of the jet nozzle **110** may itself be selectively decreased or increase instead of or in addition to the cross-sectional area $A_{sub.t}$ of a deformable throat body **120**. For example, the cross-sectional area $A_{sub.n}$ of the jet nozzle **110** may be selectively changed by translating a tapered pin axially along the nozzle **110** or pressurizing an inflatable bladder similar to the plenum **140** described above. As another possibility, two jet nozzles **110** may be used, one for achieving PEEP at low nozzle flow $V'_{sub.n}$ and the other for achieving desired $P_{sub.shutoff} - V'_{sub.aw-max}$ performance. Exemplary data of candidate jet nozzle diameters $D_{sub.t}$ for use with the disclosed embodiments is shown in Table 7, below:

(39) TABLE-US-00007 TABLE 7 $V'_{sub.n}$ (slpm): $P_{sub.n}$ 0.94 mm 1.1 mm 1.2 mm 1.4 mm 1.5 mm 1.6 mm (psig) 0.037" 0.043" 0.048" 0.055" 0.059" 0.063" 1.0 5.0 5.8 8.0 10.9 12.0 2.0 8.3 8.9 12.3 14.9 16.8 4.0 11.4 12.9 17.8 22.0 24.8 6.0 13.9 15.8 21.3 26.7 30.3 8.0 15.8 18.1 24.9 31.2 10.0 17.8 20.1 28.0 12.0 19.8 22.0 30.3 14.0 21.3 24.0 16.0 23.0 25.9 18.0 24.9 28.1 20.0 26.4 30.0 22.0 28.2 24.0 29.8 26.0 31.6 28.0 32.9 30.0 34.4

(40) For ease of explanation, the above disclosure assumes that the variable throat jet venturi **100** has a single jet nozzle **110**. As such, the cross-sectional area $A_{sub.n}$ is described as corresponding to the diameter $D_{sub.n}$ of the jet nozzle **110**. However, the disclosure is not limited in this regard. For example, the variable throat jet venturi **100** may include a plurality of jet nozzles **110** arranged in a ring or other pattern. In this case, the cross-sectional area $A_{sub.n}$ may refer to the total cross-sectional area of the plurality of jet nozzles **110** for purposes of evaluating the ratio $A_{sub.t}/A_{sub.n}$.

(41) The above description is given by way of example, and not limitation. Given the above disclosure, one skilled in the art could devise variations that are within the scope and spirit of the invention disclosed herein. Further, the various features of the embodiments disclosed herein can be used alone, or in varying combinations with each other and are not intended to be limited to the specific combination described herein. Thus, the scope of the claims is not to be limited by the illustrated embodiments.

Claims

1. A variable throat jet venturi for delivering ventilation gas to a patient, the variable throat jet venturi comprising: a jet nozzle; a deformable throat body arranged to receive ventilation gas output by the jet nozzle and defining a gas inlet and a gas outlet; and a housing containing the deformable throat body, the housing defining a pilot pressure port for pressurizing a plenum between an outer wall of the deformable throat body and an inner wall of the housing, the housing having a portion that flares outwardly away from the plenum and defines an entrainment opening which is arranged to draw ambient air directly therein, wherein the deformable throat body is arranged to receive the ventilation gas output by the jet nozzle through the entrainment opening of the housing.
2. The variable throat jet venturi of claim 1, wherein a distal tip of the jet nozzle is outside the entrainment opening.
3. The variable throat jet venturi of claim 1, wherein, with the plenum in a first pressurization state, the variable throat jet venturi achieves a shutoff pressure $P_{\text{sub.shutoff}}$ at the gas outlet of at least 55 cmH₂O at a jet nozzle pressure $P_{\text{sub.n}}$ of 10.5 psig and a jet nozzle flow $V'_{\text{sub.n}}$ equal to or less than 30 slpm and, with the plenum in a second pressurization state, the variable throat jet venturi achieves a gas outlet pressure $P_{\text{sub.aw}}$ of 10 cmH₂O at a jet nozzle flow $V'_{\text{sub.n}}$ of less than 10 slpm.
4. The variable throat jet venturi of claim 3, wherein, with the plenum in the second pressurization state, the variable throat jet venturi achieves a gas outlet pressure $P_{\text{sub.aw}}$ of 10 cmH₂O at a jet nozzle flow $V'_{\text{sub.n}}$ of less than 5 slpm.
5. The variable throat jet venturi of claim 3, wherein, with the plenum in the first pressurization state, a ratio $A_{\text{sub.t}}/A_{\text{sub.n}}$ of a cross-sectional area of the deformable throat body to a cross-sectional area of the jet nozzle is between 20 and 30 and, with the plenum in the second pressurization state, the ratio $A_{\text{sub.t}}/A_{\text{sub.n}}$ is between 2.0 and 5.0.
6. A patient ventilation interface comprising: the variable throat jet venturi of claim 1; and a nasal coupler for fluidly coupling the gas outlet of the deformable throat body to a nostril of the patient.
7. The patient ventilation interface of claim 6, wherein the nasal coupler comprises a nasal pillow.
8. A non-invasive ventilation system comprising: the patient ventilation interface of claim 6; and a pilot pressure line fluidly coupled to the pilot pressure port.
9. The non-invasive ventilation system of claim 8, further comprising a controller programmed to energize the pilot pressure line to constrict the deformable throat body during an exhalation phase of positive end-expiratory pressure (PEEP) therapy.
10. The non-invasive ventilation system of claim 8, further comprising a multi-lumen tube having a ventilation gas lumen terminating in the jet nozzle and a pilot pressure lumen in fluid communication with the pilot pressure line.
11. A method of varying a ratio between a throat diameter and a jet nozzle diameter of a variable throat jet venturi for delivering ventilation gas to a patient, the method comprising: providing a deformable throat body arranged to receive ventilation gas output by a jet nozzle, the deformable throat body defining a gas inlet and a gas outlet; and pressurizing a plenum between an outer wall of the throat body and an inner wall of a housing containing the deformable throat body to constrict the deformable throat body, the housing having a portion that flares outwardly away from the plenum and defines an entrainment opening which is arranged to draw ambient air directly therein, wherein the deformable throat body is arranged to receive the ventilation gas output by the jet nozzle through the entrainment opening of the housing.
12. The method of claim 11, wherein said pressurizing the plenum includes energizing a pilot pressure line fluidly coupled to a pilot pressure port defined by the housing.
13. The method of claim 12, wherein said energizing is performed during an exhalation phase of

positive end-expiratory pressure (PEEP) therapy.

14. The method of claim 11, wherein said pressurizing comprises pressurizing the plenum from a first pressurization state, in which the variable throat jet venturi achieves a shutoff pressure $P_{sub.shutoff}$ at the gas outlet of at least 55 cmH₂O at a jet nozzle pressure $P_{sub.n}$ of 10.5 psig and a jet nozzle flow $V'_{sub.n}$ equal to or less than 30 slpm, to a second pressurization state, in which the variable throat jet venturi achieves a gas outlet pressure $P_{sub.aw}$ of 10 cmH₂O at a jet nozzle flow $V'_{sub.n}$ of less than 10 slpm.

15. The method of claim 14, wherein, with the plenum in the second pressurization state, the variable throat jet venturi achieves a gas outlet pressure $P_{sub.aw}$ of 10 cmH₂O at a jet nozzle flow $V'_{sub.n}$ of less than 5 slpm.

16. The method of claim 14, wherein, with the plenum in the first pressurization state, a ratio $A_{sub.t}/A_{sub.n}$ of a cross-sectional area of the deformable throat body to a cross-sectional area of the jet nozzle is between 20 and 30 and, with the plenum in the second pressurization state, the ratio $A_{sub.t}/A_{sub.n}$ is between 2.0 and 5.0.

17. A variable throat jet venturi for delivering ventilation gas to a patient, the variable throat jet venturi comprising: a jet nozzle; a deformable throat body arranged to receive ventilation gas output by the jet nozzle and defining a gas inlet and a gas outlet; and a housing containing the deformable throat body, the housing defining a pilot pressure port for pressurizing a plenum between an outer wall of the deformable throat body and an inner wall of the housing, the housing having a portion that flares outwardly away from the plenum and defines an entrainment opening which is arranged to draw ambient air directly therein, wherein the deformable throat body is arranged to receive the ventilation gas output by the jet nozzle through the entrainment opening of the housing.

18. The variable throat jet venturi of claim 17, wherein, with the plenum in a first pressurization state, the variable throat jet venturi achieves a shutoff pressure $P_{sub.shutoff}$ at the gas outlet of at least 55 cmH₂O at a jet nozzle pressure $P_{sub.n}$ of 10.5 psig and a jet nozzle flow $V'_{sub.n}$ equal to or less than 30 slpm and, with the plenum in a second pressurization state, the variable throat jet venturi achieves a gas outlet pressure $P_{sub.aw}$ of 10 cmH₂O at a jet nozzle flow $V'_{sub.n}$ of less than 10 slpm.

19. The variable throat jet venturi of claim 18, wherein, with the plenum in the second pressurization state, the variable throat jet venturi achieves a gas outlet pressure $P_{sub.aw}$ of 10 cmH₂O at a jet nozzle flow $V'_{sub.n}$ of less than 5 slpm.

20. The variable throat jet venturi of claim 18, wherein, with the plenum in the first pressurization state, a ratio $A_{sub.t}/A_{sub.n}$ of a cross-sectional area of the deformable throat body to a cross-sectional area of the jet nozzle is between 20 and 30 and, with the plenum in the second pressurization state, the ratio $A_{sub.t}/A_{sub.n}$ is between 2.0 and 5.0.
